Voice Agent for Hospital Training Scenarios

Introduction and Background

Simulation-based training (SBT) has become a cornerstone of medical education, providing realistic and immersive practice environments for healthcare professionals [1]. In SBT, learners can hone both technical and communication skills in a safe, controlled setting, repeatedly practicing without risk to real patients [1]. Immediate feedback and structured debriefing in simulation further enhance learning and confidence [1]. However, traditional simulations often require specialized manikins, physical space, and instructor oversight, which can be costly and logistically challenging [2]. Recent advances in artificial intelligence (AI) and speech technology suggest new opportunities: voice-based virtual agents can simulate patient interactions, offering flexible, interactive training. Indeed, modern AI voice assistants leverage automatic speech recognition (ASR), natural language processing (NLP), and text-to-speech (TTS) to enable natural conversations [3]. In healthcare, experts foresee voice assistants being used widely as hands-free instruction tools and communication aids for medical staff [4]. By combining these technologies, a voice-driven simulation platform could bring scalable, on-demand training scenarios into hospitals and educational programs.

Problem Statement

Current clinical training methods can be limited by resource constraints and lack of realism. High-fidelity manikins and virtual reality setups are expensive and often require trained operators [2]. Moreover, existing simulations may not fully engage learners in communication or decision-making tasks, especially under stress. Role-play with standardized patients is valuable but limited by scheduling and can be inconsistent. As a result, trainees may not experience the variety of real-world scenarios needed to build confidence. The COVID-19 pandemic highlighted the need for remote and virtual training solutions, as students lost access to clinical rotations [5] [6]. Hospitals seek ways to provide repetitive, scenario-based practice (e.g. emergency codes, patient interviews) without risking patient safety or incurring high costs. A voice-based simulation agent could address these gaps by delivering realistic scenarios and immediate feedback on-demand.

Proposed Solution Overview

We propose a **Voice Agent for Hospital Training**: an AI-driven conversational simulator that emulates patient or scenario responses in real-time. Trainees interact by speaking to the agent, which understands spoken input, processes it with NLP/AI, and replies with synthesized speech. The system would run scripted scenarios (e.g. code blue, trauma triage, patient counseling) where the agent plays the role of patient or colleague. For example, a trainee might ask "What's hurting?" and the agent could describe symptoms. The system tracks the trainee's queries and actions, offering tips or scoring accuracy. By using speech interfaces rather than text, the simulation mimics real clinical communication. This approach builds on emerging tools: commercial simulators like ALEX already incorporate AI voice, answering clinical questions using speech recognition [7]. Our solution extends this by adding customizable scenario flows and automated feedback loops. Ultimately, the voice agent aims to reinforce diagnostic reasoning, communication skills, and procedural decision-making in a safe, repeatable way.

Technical Architecture and Technology Stack

The voice agent relies on a **speech-AI pipeline**: audio input → speech-to-text (ASR) → natural language understanding (NLU) → dialogue management → text-to-speech (TTS) output. For ASR, we could use services like Google Speech-to-Text, AWS Transcribe, or open-source models (e.g. Whisper) to convert the trainee's spoken words into text [8] [3]. The text is then processed by NLP components: intent and entity recognition to understand the question or command. Here, pretrained language models (such as specialized medical BERT or GPT-4) can interpret free-form medical language. Dialogue management maintains scenario state (e.g. patient vital signs, time elapsed) and determines the agent's response, potentially with the help of rule-based state machines or ML-driven policies. Responses are converted back to speech via TTS engines (e.g. Amazon Polly, Google WaveNet) for lifelike delivery [9] [3].

[8] *Figure: System architecture for the voice agent (example using Amazon Lex). Audio from the user is sent to an NLP service (ASR + intent processing). A backend logic layer (e.g. AWS Lambda) generates a response based on the scenario state (stored in a database), then returns text to a speech synthesis service.*

Key components include:
- **Audio Capture & ASR:** A microphone interface captures speech. ASR models transcribe audio in real-time. Technologies: Google Cloud Speech-to-Text, AWS Transcribe, PocketSphinx, Facebook's Wav2Vec, etc.
- **Natural Language Processing:** Interprets the transcript. This involves intent classification (e.g. asking about symptoms, ordering tests) and entity extraction (e.g. medicine names, patient attributes). Medical ontologies (like SNOMED) can help interpret domain-specific terms. We can leverage transformer-based models for flexible understanding [5].
- **Dialogue Manager / Scenario Engine:** Manages the scripted scenario flow. It keeps track of conversation history and clinical context. For example, if a trainee asks about chest pain, the manager updates the differential diagnosis state. It decides the next agent utterance and which patient responses to simulate. The manager also triggers real-time feedback: e.g. "Trainee missed a critical question" or "Good recognition of symptom".
- **Text-to-Speech (TTS):** Converts agent replies into natural-sounding voice. Services like Amazon Polly or Google Cloud TTS are suitable. They allow voice selection (male/female, accent) and have medical vocabulary support [9].
- **Backend and Data:** A database stores scenario definitions (dialogue trees, correct responses, scoring rubrics) and logs of trainee performance. An analytics module can score performance metrics (timeliness, completeness of questions, correctness of actions). We can host the system on the cloud (e.g. AWS, Azure) for scalability.

By combining these elements, the system creates a real-time feedback loop: the trainee speaks, the agent responds, and performance is evaluated on the fly [5] [3].

Implementation Details and Development Process

The project was developed iteratively over the hackathon timeline. Initially, we designed several medical scenarios (e.g. acute chest pain, severe trauma, pediatric fever) with clear learning objectives. Each scenario was mapped out as a decision flowchart to guide agent responses. We then implemented a prototype using cloud NLP APIs: for instance, a Node.js or Python client captures microphone audio and streams it to a speech-to-text API. The recognized text triggers calls to a dialogue management function (for example, an AWS Lambda or Google Cloud Function), which applies the scripted logic and formulates the next utterance. Training data for medical intents was gathered from textbooks and simulated case transcripts to fine-tune the NLU model or to define rule-based keywords. For voice output, we integrated a TTS API; initial tests used a generic voice, and we plan to refine with a clinical domain voice model. The system components were containerized (e.g. using Docker) to allow rapid deployment and scalability.

In development, we adopted agile sprints: building a minimal working loop first (strawman patient Q&A), then adding complexity (branching logic, scoring). We tested the prototype with sample dialogues and refined it based on user feedback. Clinician mentors reviewed scenarios and provided realistic dialogue lines to improve naturalness. Unit tests ensured ASR/NLP accuracy for key medical terms. By the end of the hackathon, the MVP could run a simple case end-to-end, capturing a trainee's spoken questions and providing plausible patient replies with instantaneous feedback cues (e.g. logging missed critical questions).

Visualizations and Example Scenarios

The following figures illustrate key aspects of the voice agent system:

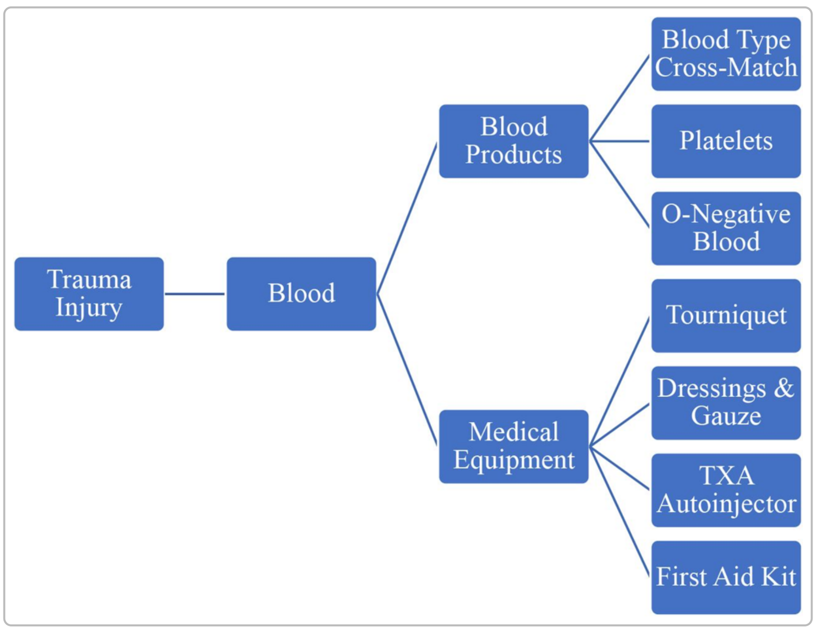

Figure 1: Example flowchart of a simulated trauma scenario. A trainee encounters a "Trauma Injury" case and must inquire about bleeding (e.g. "Does the patient have external bleeding?"). The agent's responses (e.g. "Yes, heavy external bleeding") lead the trainee through branching choices such as ordering blood products or tourniquet. This flowchart structure guides the dialogue logic in the simulation.

In Figure 1, the simulation flows from an initial emergency presentation ("Trauma Injury") through decision nodes (ordering blood, addressing airway, etc.). At each branch, the agent's voice responses depend on the trainee's questions and actions. Such flowcharts are implemented in the dialogue manager to determine what the agent says next and what feedback to provide. For instance, if the trainee fails to control hemorrhage, the agent might prompt with “The patient’s blood pressure is dropping.”

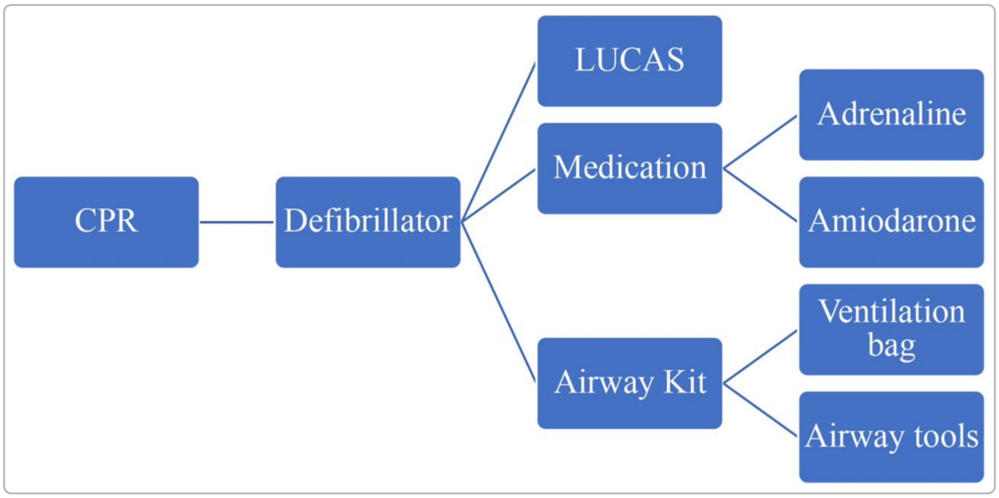

Figure 2: Example flowchart of a CPR training scenario. Here, a “CPR” event leads the trainee to use a defibrillator and airway equipment. The agent (acting as the patient) might respond to questions about pulse or breathing. Branches include actions like administering medications (e.g. adrenaline) or using the LUCAS chest compression device. This diagram visualizes the structured dialogue paths for the scenario.

Figure 2 depicts a cardiac arrest scenario where trainees must apply advanced life support steps. The voice agent can simulate a patient’s verbal cues (e.g. gasping for air) or interact with equipment (e.g. “The LUCAS is now active”). By following this dialogue flow, the system emulates a realistic resuscitation, checking off critical actions.

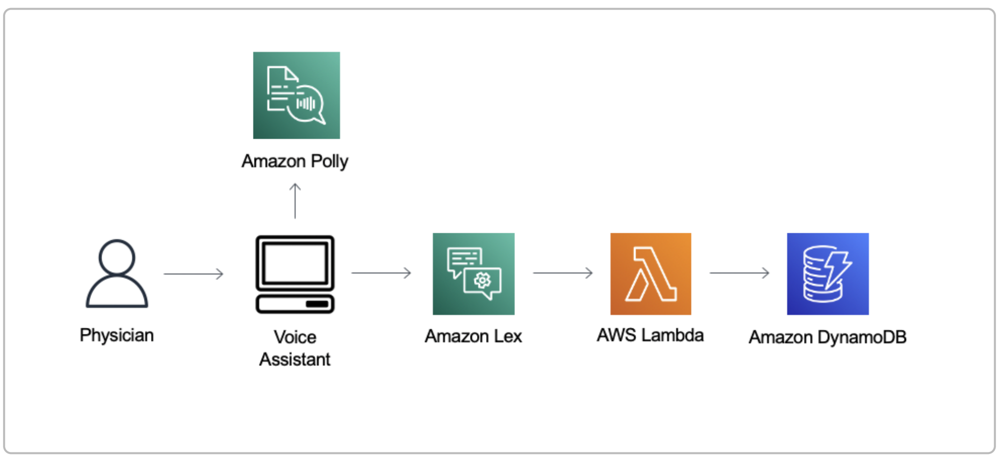

Figure 3: System architecture diagram for the voice agent (example using Amazon Lex). The trainee’s voice is captured by a microphone and sent to a voice assistant engine (e.g. Amazon Lex). An AWS Lambda function or similar processes the intent and retrieves scenario data from a database. The agent’s text response is sent to Amazon Polly (TTS) and played back. This pipeline (ASR → NLP → Logic → TTS) forms the real-time feedback loop in our system [8].

Figure 3 illustrates a high-level architecture. The user's speech activates the voice assistant, which streams audio to an NLP service. The backend logic (cloud functions) handles the simulation state and scoring. Finally, the response text is synthesized into speech. We implemented similar pipelines using both AWS and open-source tools, ensuring HIPAA-compliant, secure data handling. The modular design allows swapping components (e.g. using Google Dialogflow instead of Lex) as needed.

Sample Dialogue (Trainee-Agent Interaction): Below is an illustrative excerpt of how a conversation might proceed in a training scenario:

- **Trainee:** “Patient is complaining of chest pain.”
- **Agent (patient):** *groans* “It hurts here…”
- **Trainee:** “Is the pain sharp or dull?”
- **Agent:** “Sharp, like someone stabbing me.”
- **Trainee:** “Do you have any shortness of breath or nausea?”
- **Agent:** “Yes, I feel really short of breath.”
- **Trainee:** “Are you sweating?”
- **Agent:** “Yes, and I feel dizzy.”
- **Trainee:** *Orders an ECG and administers nitroglycerin*
- **Agent:** “(after nitroglycerin) The pain is easing slightly, but I still feel weak.”
- **Agent (feedback):** “Good job – consider transferring to the cath lab.”

In this flow, the trainee's verbal queries drive the scenario. The agent's speech responses adapt based on supplied vitals and actions. After completion, the system provides feedback such as missed questions or errors (not shown above). This example demonstrates the natural language interaction and decision prompts offered by the voice agent.

Results or Projected Outcomes

Our prototype suggests that a voice-agent simulation could significantly enhance training efficiency and effectiveness. By leveraging NLP, the system can provide immediate, personalized feedback – a feature shown to improve learning outcomes. For instance, a similar NLP-driven simulator “Hepius” yielded a statistically significant increase in medical students' test scores after use [5]. We expect trainees using our voice agent to benefit in several ways:
- **Improved Skill Retention:** Repeated practice of scenarios (e.g. cardiac arrest codes, emergency triage) in a low-stakes environment helps cement decision-making pathways [1] [6].
- **Enhanced Communication Abilities:** Speaking with the agent improves patient interviewing and team communication skills, which are critical but harder to teach with mannequins. Real-time correction of communication missteps could be implemented later.
- **Standardization of Training:** Every trainee can experience the same evidence-based scenarios with automated scoring, ensuring consistency in evaluation. In contrast, instructor-led cases can vary.
- **Resource Savings:** Hospitals may reduce the need for physical labs and instructor time. The system runs on common devices (tablets or laptops) and requires only voice hardware, making it scalable.
- **Better Preparedness:** According to simulation experts, exposing learners to uncommon crises through simulated practice “bridges the gap between classroom and real-world” and accelerates expertise [6]. By making more scenarios accessible (including on-demand practice), hospitals can raise overall competency.

Potential outcomes include higher pass rates on practical exams, faster response times in simulated codes, and ultimately improved patient safety. Future trials could quantitatively measure such gains.

Conclusion and Future Scope

In summary, a voice-based training agent offers a promising new modality for medical education. By integrating speech interfaces, NLP, and real-time feedback, our system creates dynamic simulations that mimic real clinical interactions [5] [8]. Hospitals and educators could use this tool to supplement existing training, especially when instructor resources or physical simulation centers are limited. Looking ahead, we envision enhancements such as: expanding scenario libraries (e.g. pediatric cases, diagnostic puzzles), supporting multilingual voice agents, and incorporating advanced LLMs for even more natural dialogue. Integration with VR or mannequin simulators is also possible, adding voice layers to physical scenarios. Ongoing development will focus on refining the NLP for medical accuracy, measuring learning outcomes in controlled studies, and ensuring data privacy. With further research and iteration, voice agents could become a standard feature of healthcare training, improving preparedness and patient outcomes.

References

- Barra et al., *Resuscitation* (2024). Shaping the future of simulator interactions: the role of ChatGPT's Advanced Voice Mode [4].
- Edwards et al., *PSNet Patient Safety Primer* (2023). Simulation-based training in healthcare [6].
- Furlan et al., *JMIR Med Inform* (2021). A natural language processing–based virtual patient simulator for clinical diagnostic reasoning [5].
- Ismail et al., *BMC Med Educ* (2024). Simulation challenges in healthcare education (developing world) [10].
- AWS Machine Learning Blog (2021). Custom voice assistant using Amazon Lex [8].
- Simulaids (2025). **ALEX Plus** Patient Simulator – AI voice responses and speech recognition [7].
- Ermolina et al., *JMIR Form Res* (2021). Voice-controlled intelligent personal assistants in healthcare (Delphi study) [4].

---

[1] The impact of simulation-based training in medical education: A review - PMC
https://pmc.ncbi.nlm.nih.gov/articles/PMC11224887/

[2] [10] Challenges and opportunities in the uptake of simulation in healthcare education in the developing world: a scoping review - PMC
https://pmc.ncbi.nlm.nih.gov/articles/PMC11384200/

[3] Gladia - Building AI Voice Agents: Starter Guide
https://www.gladia.io/blog/building-ai-voice-agents-starter-guide

[4] Voice-Controlled Intelligent Personal Assistants in Health Care: International Delphi Study - PMC
https://pmc.ncbi.nlm.nih.gov/articles/PMC8065565/

[5] JMIR Medical Informatics - A Natural Language Processing–Based Virtual Patient Simulator and Intelligent Tutoring System for the Clinical Diagnostic Process: Simulator Development and Case Study
https://medinform.jmir.org/2021/4/e24073/

[6] Simulation Training | PSNet
https://psnet.ahrq.gov/primer/simulation-training

[7] Simulaids ALEX Plus Patient Simulator with Speech Recognition Ability
https://www.universalmedicalinc.com/simulaids-alex-plus-patient-simulator.html?srsltid=AfmBOoqqqoVg_MVm-EJEK98W88PQVQ22m9mwjD7t6p4lIL7hjBTR_w5a

[8] [9] Simplify patient care with a custom voice assistant using Amazon Lex V2 | AWS Machine Learning Blog
https://aws.amazon.com/blogs/machine-learning/simplify-patient-care-with-a-custom-voice-assistant-using-amazon-lex-v2/